



SUPPLY CHAIN DELIVERY PERFORMANCE REPORT

End-to-End Order Fulfillment Analysis & Predictive Insight

172,765 Orders Analyzed | Global E-Commerce Operations | 2026



. TABLE OF CONTENTS

- 1 Executive Summary
 - 2 Business Context & Objectives
 - 3 Key Performance Indicators (KPIs)
 - 4 Profitability Analysis
 - 5 Bottleneck Detection
 - 6 Root Cause Analysis
 - 7 Time-Based Delay Patterns
 - 8 Machine Learning Model Results
 - 9 Strategic Recommendations
 - 10 Conclusion
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1. EXECUTIVE SUMMARY

This report presents a comprehensive analysis of the delivery operations of a global e-commerce company managing end-to-end order fulfillment across multiple regions. The analysis covers 172,765 orders, quantifying their financial impact and establishing a data-driven framework for improvement.

The headline finding is stark: 54.71% of all orders are delivered late, costing the business \$2.1M in eroded profitability on delayed orders alone. While total profit across fulfilled orders reached \$7.5M, the persistent late-delivery problem represents both a significant financial drain and a major customer experience risk. A Random Forest predictive model achieved 74% accuracy in identifying at-risk orders before they become late.

Key Finding	Metric
Overall Late Delivery Rate	54.71%
On-Time Delivery Rate	45.29%
Total Orders Analyzed	172,765
Total Profit (profitable orders)	\$7.5M
Profit at Risk (delayed orders)	\$2.1M
Predictive Model Accuracy (RF)	74%



2. BUSINESS CONTEXT & OBJECTIVES

The company operates a global e-commerce platform selling products across categories including sporting goods, fitness equipment, outdoor gear, footwear, and apparel across multiple international regions.

Core Business Problem

Actual shipping times frequently deviate from scheduled delivery windows, creating eroded customer trust, reduced order profitability, and an inability to make reliable commitments to buyers at the point of purchase.

Analytical Objectives

- ▶ Understand the current state of delivery performance across all dimensions (region, mode, time, segment)
- ▶ Quantify the financial impact of delays on order profitability
- ▶ Identify the primary operational bottlenecks driving late deliveries
- ▶ Build a predictive model to flag high-risk orders before they are shipped
- ▶ Deliver actionable recommendations to improve on-time delivery and profitability



3. KEY PERFORMANCE INDICATORS (KPIs)

The following KPIs were computed directly from the cleaned dataset to establish a performance baseline.

172,765

Total Orders

94,523

Late Deliveries

54.71%

Late Delivery %

45.29%

On-Time Delivery %

\$7.5M

Total Profit

\$2.1M

Profit at Risk

3 days

90th Pctl. Delay

\$22.03

Avg. Order Profit

**LATE DELIVERY RATE: 54.71%**

More than half of all orders arrive late. This is not an edge-case problem — it is the default experience for the majority of customers.

**\$2.1M PROFIT AT RISK**

Orders that experienced delays collectively generated \$2.1M in profit that is under constant pressure from further operational deterioration.

**90TH PERCENTILE DELAY = 3 DAYS**

Even the most extreme cases are contained within 3 days of lateness, suggesting the problem is systemic and process-driven rather than catastrophic.



4. PROFITABILITY ANALYSIS

4.1 Profitability Distribution

Order-level profitability was classified into three tiers based on Order Profit Per Order. While 80.7% of orders are profitable, the 18.7% loss-making share represents a meaningful drag that is disproportionately concentrated among delayed shipments.

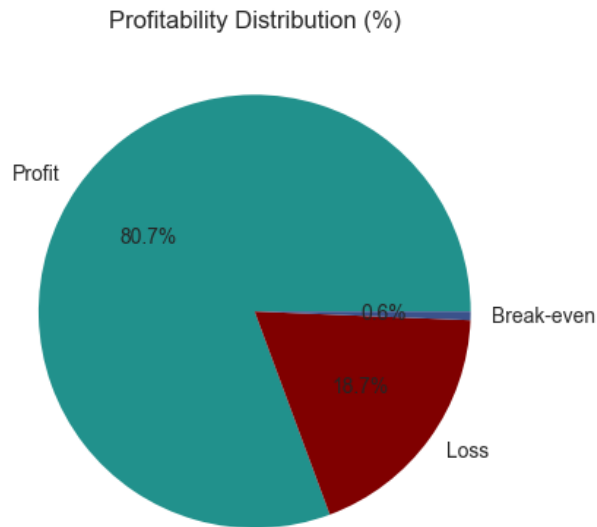


Figure 4.1 — Profitability Distribution by Order Tier



KEY INSIGHT

80.7% of orders are profitable, but the 18.7% loss-making segment is concentrated among delayed shipments — linking delivery performance directly to margin erosion.

4.2 Delay Distribution & Profit vs. Delay Days

The delay distribution shows that 31.0% of all orders arrive exactly 1 day late — the single largest cohort. Combined, orders delayed by 1–4 days account for 54.7% of all order volume, directly mapping to the overall late delivery rate.

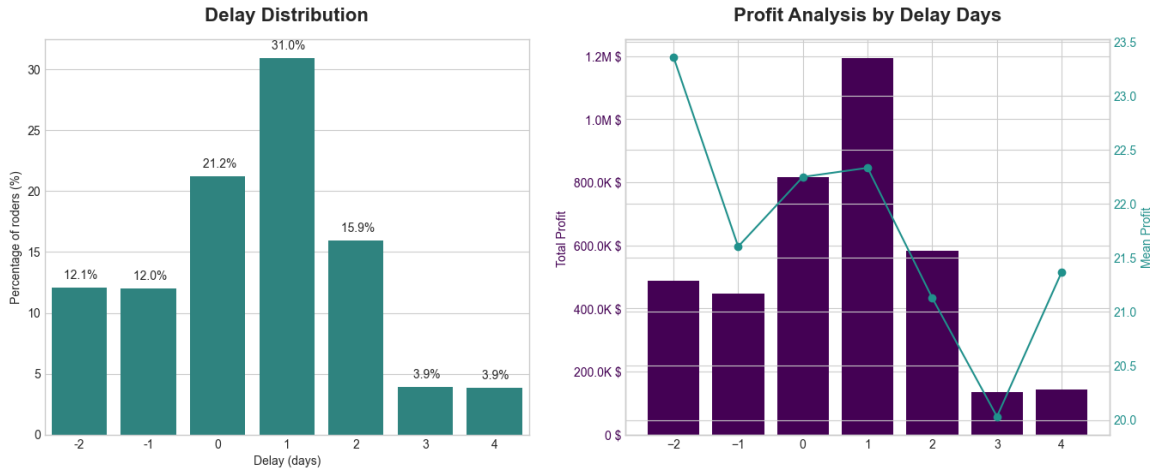


Figure 4.2 — Delay Distribution & Profit vs. Delay Days

A critical observation: mean profit per order remains stable at approximately \$21–\$23 across all delay levels. The profitability problem is therefore driven by volume of delayed orders, not individual order economics — making systemic throughput improvement the highest-leverage intervention.

KEY INSIGHT

31% of delayed orders are exactly one day late, suggesting process inefficiencies rather than severe disruptions. Fixing the last mile of on-time dispatch would resolve the majority of the problem.



5. BOTTLENECK DETECTION

Delay percentage was computed across six key operational dimensions. The charts below reveal where the fulfillment process breaks down most severely.

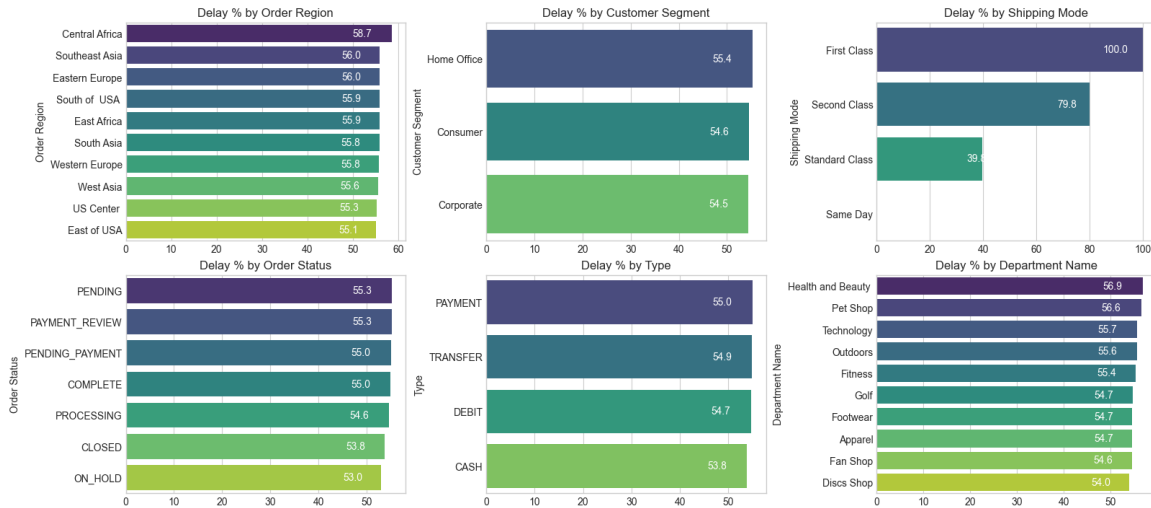


Figure 5 — Delay Rate by Operational Dimension



SHIPPING MODE IS THE #1 LEVER

First Class: 100% delay rate. Second Class: 79.8%. Standard Class: 39.8%. Same Day: 0%. Shipping mode assignment logic is the single most impactful lever to fix.



REGIONAL SPREAD IS NARROW (55–59%)

Central Africa leads at 58.7%, but all regions cluster between 55–59%. This rules out a localized logistics failure and confirms a company-wide systemic issue.



CUSTOMER SEGMENTS NEARLY IDENTICAL (54.5–55.4%)

No segment receives preferential service. Consumer, Corporate, and Home Office customers all experience the same broken delivery promise.



HEALTH & BEAUTY AND PET SHOP LEAD BY DEPARTMENT

Health & Beauty (56.9%) and Pet Shop (56.6%) marginally outpace other departments, warranting an inventory and carrier audit for these categories.

6. ROOT CAUSE ANALYSIS

Central Africa was selected for deep-dive root cause analysis as the highest-delay region (58.7%). The chart below ranks the top 10 driver factors by delay percentage within this region.

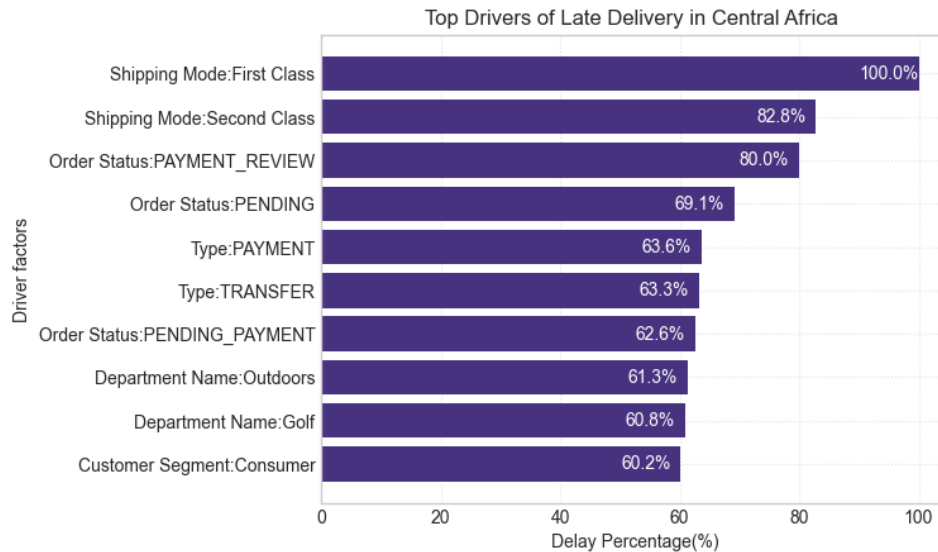


Figure 6 — Top 10 Delay Drivers, Central Africa

KEY INSIGHT

Payment review and pending-payment statuses rank among the top delay drivers in Central Africa, pointing to process friction upstream of shipping itself.

7. TIME-BASED DELAY PATTERNS

Three temporal dimensions were analyzed: month of year, day of week, and hour of day. While delay rates are relatively stable across all dimensions (53–57%), specific peaks highlight opportunities for targeted capacity planning.

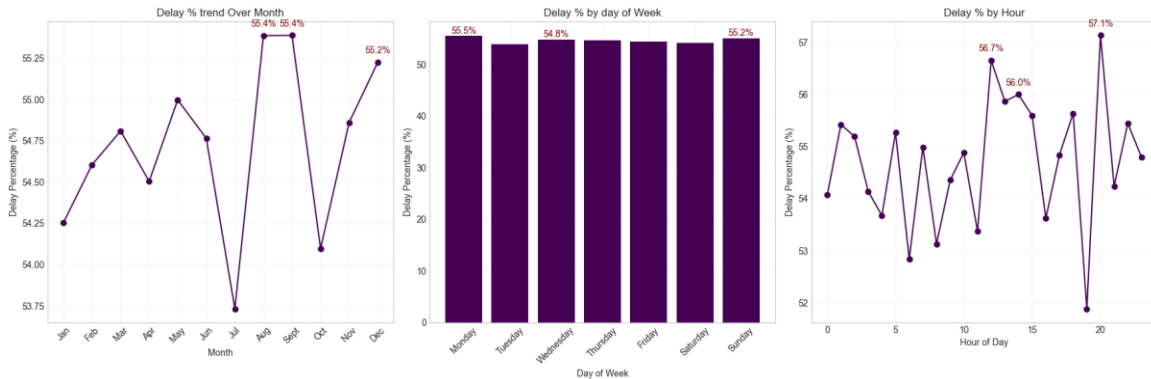


Figure 7 — Delay Rate by Month, Day of Week, and Hour of Day

💡 PEAK DELAY MONTHS: AUG, SEPT & DEC

Aug and Sept are the highest months at 55.4%, with Dec close behind — reflecting mid-year promotions and Q4 holiday surge overwhelming fulfillment capacity. July is a notable low point (53.75%), confirming seasonal variation is plannable.

💡 DAY-OF-WEEK VARIANCE IS MINIMAL

Monday (55.5%) and Sunday (55.2%) are the peak days, but the spread across the week is less than 1 percentage point — not a meaningful lever for operational intervention.

💡 INTRA-DAY PEAKS: HOUR 21, 11 & 12

Late-evening orders (hour 21) have the highest delay rate at 57.1%, likely reflecting processing cutoff windows. Midday peaks suggest bottlenecks during peak-volume hours.



8. MACHINE LEARNING MODEL RESULTS

A supervised classification model was built to predict whether an individual order will be delivered late (Late_delivery_risk = 1). The pipeline included frequency encoding of categorical variables, a stratified train/test split, SMOTE oversampling to address class imbalance, and Random Forest classification.

Class Imbalance Handling (SMOTE)

Split	Class 0 (On-Time)	Class 1 (Late)
Before SMOTE (train)	59,030	79,182
After SMOTE (train)	79,182	79,182

Random Forest Classifier — Test Set Performance (34,553 records)

Metric	Class 0 (On-Time)	Class 1 (Late)	Overall (Weighted)
Precision	0.68	0.78	0.74
Recall	0.72	0.75	0.74
F1-Score	0.70	0.77	0.74
Accuracy	—	—	0.74
Support	14,758	19,795	34,553



74% OVERALL ACCURACY

The model correctly classifies nearly 3 in 4 orders — a strong foundation for production deployment as an order-level late-delivery alert system.



HIGHER PRECISION ON LATE ORDERS (0.78)

When the model predicts an order will be late, it is correct 78% of the time — sufficient to trigger targeted interventions without generating excessive false alarms.

 RECALL FOR LATE ORDERS AT 0.75

The model captures 75% of actual late orders. The 25% miss rate is acceptable for a first-generation alert system, with clear improvement potential.

 CLASS 0 HARDER TO PREDICT (PRECISION 0.68)

On-time deliveries share many characteristics with late ones given the systemic nature of delays, making them genuinely difficult to distinguish.

★ 9. STRATEGIC RECOMMENDATIONS

Based on the findings across all analytical dimensions, the following prioritized actions are recommended:

Priority	Action	Expected Impact
Critical	Audit First & Second Class Shipping Capacity First Class runs a 100% delay rate, Second Class 79.8%. Conduct an immediate carrier SLA audit; consider suspending or repricing First Class until resolved.	Very High
High	Deploy the Predictive Alert System The Random Forest model (74% accuracy, 78% precision on late orders) is ready for production piloting to flag high-risk orders at confirmation.	High
High	Resolve Payment Processing Bottlenecks PAYMENT_REVIEW (55.3% overall; 80.0% in Central Africa) and PENDING_PAYMENT (55.0%) show elevated delay rates. Automate escalation past 2 hours.	High
Medium	Develop Seasonal Surge Capacity Plans Aug, Sept (55.4%) and Dec (55.2%) are peak delay months. Build pre-season capacity, temporary logistics partnerships, and adjusted delivery windows.	Medium
Medium	Default to Standard Class for Eligible Orders Standard Class achieves a 39.8% delay rate — far superior to First and Second Class. Re-evaluate shipping mode assignment logic.	Medium
Medium	Investigate High-Delay Departments in Africa Outdoors (61.3%) and Golf (60.8%) in Central Africa show elevated delay rates. Audit stock availability, pick-and-pack times, and carrier assignment.	Medium
Low	Reduce Loss-Making Order Share (18.7%) Warrants a separate pricing and discount review; combine with delay analysis to identify compounded margin risk.	Long-term
Low	Continuously Retrain the Predictive Model Retrain quarterly; add carrier-level, weather-event, and warehouse-utilization features. Target recall above 80% within 6 months.	Long-term



10. CONCLUSION

This analysis has surfaced a clear and urgent picture: a global e-commerce operation where the majority of orders (54.71%) fail to meet their promised delivery windows, costing \$2.1M in at-risk profit and undermining customer trust at scale.

The root causes are identifiable and addressable. Shipping mode misconfiguration (First Class at 100% late, Second Class at 79.8%) is the dominant operational failure. Payment processing friction creates a secondary bottleneck. Seasonal volume spikes are not adequately planned for. Geographic disparities, while present, are secondary to the systemic company-wide pattern.

A Random Forest predictive model trained on readily available order features already achieves 74% accuracy in identifying at-risk orders. This is a deployable tool, not a future aspiration.

Path to Target State

Priority Area	Current State	Target State
Late Delivery Rate	54.71%	< 30% within 12 months
First Class On-Time Rate	0%	> 80%
Second Class On-Time Rate	20.2%	> 60%
Predictive Model Accuracy	74%	> 82% (enhanced model)
Loss-Making Orders	18.7%	< 12%
Profit at Risk	\$2.1M	Reduce by 40%

With focused execution on the eight recommendations in this report — starting with the critical shipping mode audit and predictive alert system deployment — the business can systematically recover its delivery performance, protect its \$7.5M profit base, and build the operational reliability that sustained e-commerce growth demands.